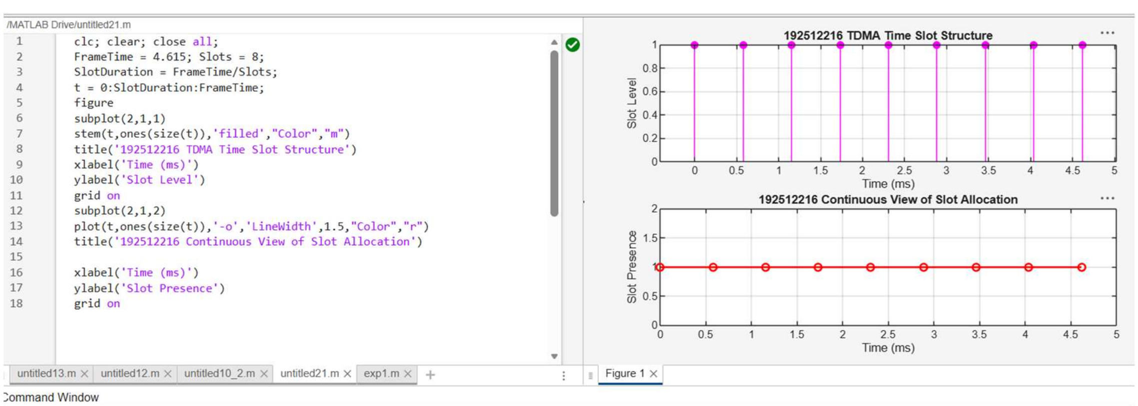
Ex No. 2: Implementing TDMA Slot Allocation in GSM

AIM: To implement TDMA (Time Division Multiple Access) slot allocation in GSM using MATLAB and analyze system performance in terms of time slot duration, frame utilization, and number of allocated slots.

Objective 1: Matlab Code and Output

```
clc; clear; close all;
FrameTime = 4.615;
Slots = 8; SlotDuration =
FrameTime/Slots; t =
0:SlotDuration:FrameTime; figure
subplot(2,1,1) stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)') ylabel('Slot Level')
grid on subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')
xlabel('Time (ms)') ylabel('Slot Presence')
grid on
```



Objective 2:Matlab code and output

clc;

clear; close all;

TotalSlots= 8;

AllocatedSlots = 6;

FreeSlots = TotalSlots - AllocatedSlots;

Utilization = (AllocatedSlots/TotalSlots)*100;

disp('Frame Utilization (%)') disp(Utilization)

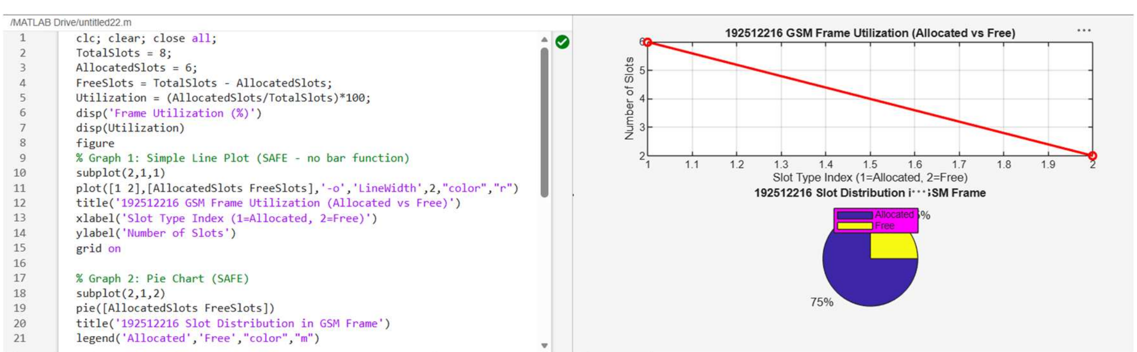
figure

% Graph 1: Simple Line Plot (SAFE - no bar function)

subplot(2,1,1) plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2) title('GSM Frame Utilization (Allocated vs Free)') xlabel('Slot Type Index (1=Allocated, 2=Free)') ylabel('Number of Slots') grid on

% Graph 2: Pie Chart (SAFE) subplot(2,1,2) pie([AllocatedSlots FreeSlots]) title('Slot Distribution in GSM Frame')

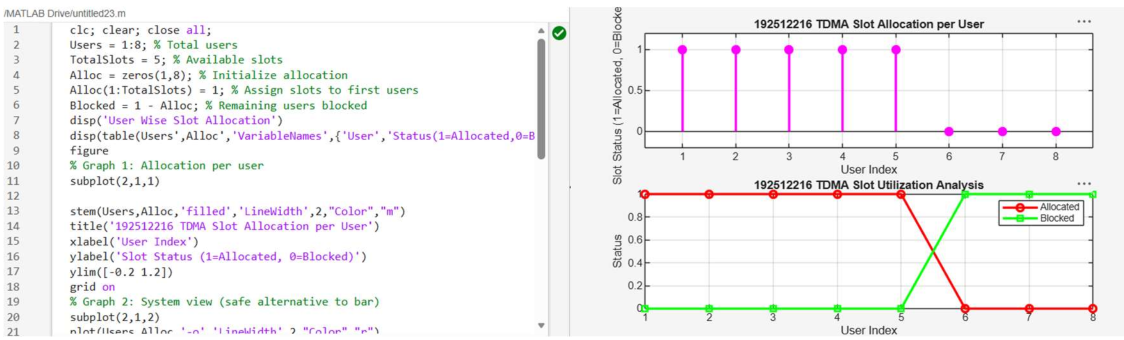
legend('Allocated','Free')



Objective 3: Matlab Code and Output

```
clc; clear; close all;

Users = 1:8; % Total users

TotalSlots = 5; % Available slots

Alloc = zeros(1,8); % Initialize allocation

Alloc(1:TotalSlots) = 1; % Assign slots to first users

Blocked = 1 - Alloc; % Remaining users blocked

disp('User Wise Slot Allocation')

disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'}
))

figure

% Graph 1: Allocation per user

subplot(2,1,1)

stem(Users,Alloc,'filled','LineWidth',2)

title('TDMA Slot Allocation per User')

xlabel('User Index')

ylabel('Slot Status (1=Allocated, 0=Blocked)')

ylim([-0.2 1.2])

grid on

% Graph 2: System view (safe alternative to bar)

subplot(2,1,2)

plot(Users,Alloc,'-o','LineWidth',2)

hold on
```

```
plot(Users,Blocked,'-s','LineWidth',2)

title('TDMA Slot Utilization Analysis')

xlabel('User Index')

ylabel('Status')

legend('Allocated','Blocked')

grid on
```



Result:

1. The time slot duration in the GSM TDMA frame was successfully calculated, confirming equal division of the frame into fixed time intervals.
2. The frame utilization was effectively analyzed, showing the proportion of allocated and free slots in the TDMA system.
3. The slot allocation per user was successfully implemented, demonstrating that limited slots are assigned to users while excess users are blocked due to resource constraints.